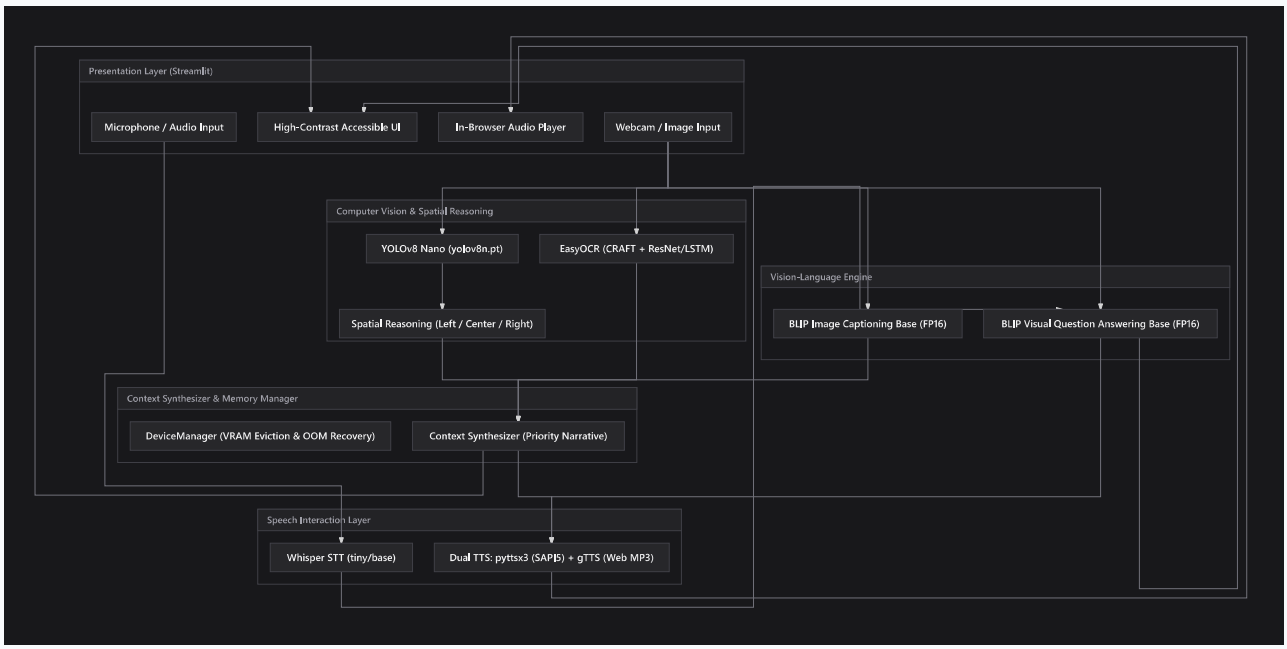
Walkthrough: AI-Powered Visual Assistant for Visually Impaired People

Executive Summary

We have successfully developed, tested, and validated the complete academic project **AI-Powered Visual Assistant for Visually Impaired People Using Computer Vision and NLP**.

The system is fully modular, runs locally on Windows with NVIDIA GeForce RTX 2050 (4 GB) GPU acceleration, enforces strict VRAM safety and dynamic model unloading, requires zero paid APIs, and provides an accessible Streamlit web application.

1. Complete Architecture & Modular Implementation



Module Breakdown:

File	Component	Description
	Streamlit Web UI	High-contrast, screen-reader friendly interface with camera capture, voice recording, live VRAM monitoring, and audio narration.
	YOLOv8 Detector	Detects 80 everyday objects, calculates spatial positions ("*on your left*", "*directly in front*", "*on your right*"), and flags immediate obstacles based on area percentage.
	EasyOCR Reader	Transcribes printed notices, signs, and labels; clusters bounding boxes in natural reading order (top-to-bottom, left-to-right).
	BLIP Captioner	Generates contextual, natural language descriptions of the camera view in FP16 precision.
	BLIP VQA Engine	Answers open-ended visual questions (e.g. "*What color is the bus?*" → "*blue*").
	Whisper STT	Transcribes spoken voice queries into text using pure-Python `soundfile` decoding (no external `ffmpeg` binary needed).
	Dual TTS Synthesizer	Offline Windows SAPI5 (`pyttsx3`) for laptop speakers + in-memory MP3 streaming (`gTTS`) for web browser playback.
	Context Synthesizer	Assembles detections, OCR text, and captions into a prioritized spoken summary (Obstacles → Scene View → Objects → Text).
	Memory & Device Guard	Inspects GPU VRAM, coordinates cache evictions (`empty_cache`), and provides automatic CPU fallback on CUDA OOM.

2. Test Verification Across All 6 Phases

All **12 unit tests** pass with 100% success rate:

```
powershell
.\.venv\Scripts\python.exe -m pytest tests/ -v
```

```
text
tests/test_detector.py::test_spatial_position_logic PASSED               [  8%]
tests/test_detector.py::test_detector_inference_on_synthetic_image PASSED [ 16%]
tests/test_detector.py::test_detector_cpu_fallback PASSED                [ 25%]
tests/test_nlp.py::test_context_synthesizer_prioritization PASSED        [ 33%]
tests/test_nlp.py::test_scene_captioner_inference PASSED                 [ 41%]
tests/test_nlp.py::test_vqa_engine_inference PASSED                      [ 50%]
tests/test_ocr.py::test_spatial_tagging_for_text PASSED                  [ 58%]
tests/test_ocr.py::test_reading_order_sorting PASSED                     [ 66%]
tests/test_ocr.py::test_ocr_inference_on_synthetic_sign PASSED           [ 75%]
tests/test_speech.py::test_tts_audio_bytes_generation PASSED             [ 83%]
tests/test_speech.py::test_tts_save_audio PASSED                         [ 91%]
tests/test_speech.py::test_speech_recognizer_lifecycle_and_transcribe PASSED [100%]
============================== 12 passed in 54.27s ==============================
```

3. How to Launch and Test the System

Option A: Launch the Accessible Web Application

```
powershell
.\.venv\Scripts\python.exe -m streamlit run app.py
```

Open `http://localhost:8501` in your browser.

Features in the Web App:

- **Full Scene Analysis**: Capture from webcam or upload an image, and click "*Describe My Surroundings*" to hear the complete spoken narrative with highlighted visual boxes.
- **Voice & Visual Q&A**: Record your voice asking a question about the view, see Whisper transcribe it, and hear BLIP VQA answer it out loud.
- **Dedicated Object Detector**: Interactive slider for detection confidence and spatial warnings.
- **Dedicated OCR Reader**: High-contrast text extraction from documents, signs, and labels.
- **Live Diagnostics Sidebar**: Displays live GPU VRAM usage and one-click VRAM clearing.

Option B: Run Standalone CLI Demos

```
powershell
# 1. YOLO Detection with Spatial Reasoning
.\.venv\Scripts\python.exe scripts/run_detector_demo.py sample_images/bus.jpg

# 2. OCR Text Extraction in Natural Reading Order
.\.venv\Scripts\python.exe scripts/run_ocr_demo.py

# 3. BLIP Scene Captioning & Visual Q&A
.\.venv\Scripts\python.exe scripts/run_vlm_demo.py sample_images/bus.jpg

# 4. Speech Recognition & Text-to-Speech
.\.venv\Scripts\python.exe scripts/run_speech_demo.py
```

4. Academic Documentation & Project Artifacts

- **Project README**: contains the full academic overview, architecture diagram, model descriptions, hardware constraints, installation instructions, and ethical disclaimers.
- **Dependencies**: contains verified version constraints for Python 3.12 and CUDA 12.4.
- **Git Protection**: excludes virtual environments, weights, caches, and temporary files.
- **Models Cache**: Organized under with separate directories for `yolo/`, `easyocr/`, `huggingface/`, and `whisper/`.